Project Sprint #1

The Solitaire game is described in CS449HomeworkOverview. You should read the description very carefully.

In this assignment, you aim to specify the requirements (i.e., user stories and acceptance criteria) of the target software that allows a human player to play a Solitaire game. These requirements will be fully implemented by the end of sprint 3. The minimum features include **choosing the board size and type, starting a new game**, **making a move, determining if a game is over**. The following is a sample GUI layout.

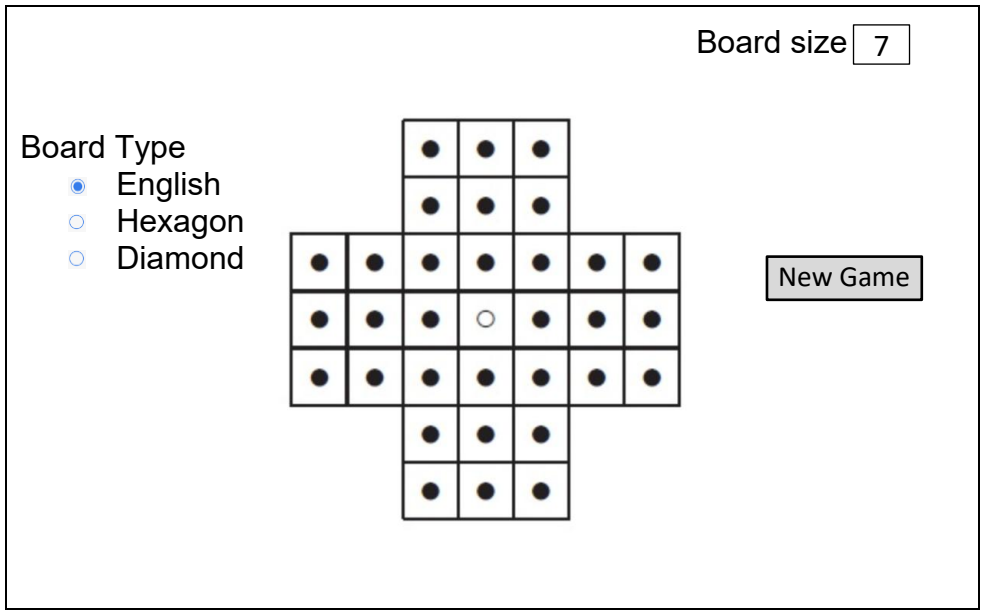


Figure 1. Sample GUI layout of the first working program by the end of Sprint 3

Use the following tables to document your user stories and acceptance criteria.

You are required to use the free ChatGPT or another LLM to complete two user stories and two of their respective acceptance criteria. You must ensure that the generated user stories are correct and fix them if not. At the end of the submission, provide screenshots of your LLM prompts and answers, along with the description of errors the LLM made that you had to correct. Points will be deducted if no screenshots/description are provided.

For the other user stories and acceptance criteria, you are not required to use LLM.

I. User Stories (3 points)

- **User Story Template**: As a <role>, I want <goal> [so that <benefit>]

ID	User Story Name	User Story Description	Priority	Estimated effort (hours)
1	Choose a board size	As a human player, I want to choose the size of the Solitaire board so that I can play the game using my preferred board size.	High	1 hr
2	Choose the board type	As a human player, I want to choose the type of Solitaire board so that I can play using my preferred board configuration.	High	1 hr

ID	User Story Name	User Story Description	Priority	Estimated effort (hours)
3	Start a new game of the chosen board size and type	As a player, I want to start a new game after choosing my desired board size and type, so that I can play using my preferred configuration.	High	2 hr
4	Make a move in a game	As a player, I want to choose what move I make in the game and have the application execute that move, so that the game progresses according to my individual choices.	High	3 hr
5	A game is over	As a player, I want the application to determine when my game is over, so that I know when I have completed the game and how well I performed.	High	1 hr

II. Acceptance Criteria (AC) (12 points): Add/delete rows as needed.

User Story ID and Name	AC ID	Description of Acceptance Criterion	Status (completed, toDo, inProgress)
1. Choose a board size	1.1	AC 1.1 – Select a valid board size Given the Solitaire application is open and the available board sizes are displayed, When the player selects one of the available board sizes, Then the selected board size is stored as the board size to be used when starting a new game.	toDo
	1.2	AC 1.2 – Change the selected board size Given the player has already selected a board size, When the player selects a different available board size, Then the newly selected board size replaces the previously selected board size.	toDo
2. Choose the board type	2.1	AC 2.1 – Select a board type Given the Solitaire application is open and the available board types are displayed, When the player selects English, Hexagon, or Diamond, Then the selected board type is stored as a board type to be used when starting a new game.	toDo
	2.2	AC 2.2 – Change the selected board type Given the player has already selected a board type, When the player selects a different available board type, Then the newly selected board type replaces the previously selected board type.	toDo
3. Start a new game of the	3.1	AC 3.1 – Start with selected configuration Given the player has selected a valid board size and board type,	toDo

User Story ID and Name	AC ID	Description of Acceptance Criterion	Status (completed, toDo, inProgress)
chosen board size and type		When the player starts a new game, Then the application creates a new game using the selected board size and board type.	
	3.2	AC 3.4 – Initialize the board Given the player has selected a valid board size and board type, When the player starts a new game, Then the application displays the selected board with all valid positions filled with marbles except for the center position, which will be empty.	toDo
	3.3	AC 3.3 – Starting another new game Given the player has chosen to play again and has selected a board size and board type, When the player starts the new game, Then the application starts a new game using the player's specified configuration.	toDo
4. Make a move in a game	4.1	AC 4.1 – Execute a valid move Given a game is currently in progress, When the player selects a valid move, Then the application executes the selected move and updates the board to reflect the player's choice.	toDo
	4.2	AC 4.2 – Reject an invalid move Given a game is currently in progress, When the player attempts a move that is not allowed, Then the application does not execute the move and the board remains unchanged.	toDo
	4.3	AC 4.3 –Execute the selected move Given a game is currently in progress and a valid move is available, When the player moves a marble orthogonally or diagonally over an adjacent marble into an empty position two spaces away, Then the application moves the selected marble to the empty position and removes the marble that was jumped over.	toDo
5. A game is over	5.1	AC 5.1 – Game ends with one marble remaining	toDo

User Story ID and Name	AC ID	Description of Acceptance Criterion	Status (completed, toDo, inProgress)
		Given a game is currently in progress, When the player completes a valid move that leaves only one marble on the board, Then the application determines that the game is over.	
	5.2	AC 5.2 - Game ends when no valid moves remain Given a game is currently in progress, When there are no valid moves remaining on the board, Then the application automatically determines that the game is over.	toDo
	5.3	AC 5.3 – Display game rating Given the application has determined that the game is over, When the final number of marbles is calculated, Then the application displays the player's rating as Outstanding for 1 marble remaining, Very Good for 2 marbles, Good for 3 marbles, or Average for 4 or more marbles.	toDo
	5.4	AC 5.4 – Prompt the player to play again Given the current game has ended, When the application displays the result of the game, Then the application prompts the player to choose whether they want to play another game.	toDo
	5.5	AC 5.5 – Player chooses to play again Given the current game has ended and the application asks whether the player wants to play again, When the player chooses to play another game, Then the application returns to the main screen where the player can select a board size and board type before starting a new game.	toDo
	5.6	AC 5.6 – Player chooses not to play again Given the current game has ended and the application asks whether the player wants to play again, When the player chooses not to play another game, Then the application exits safely.	toDo

Screenshots of LLM usage:

Prompt: I uploaded the assignment requirements and asked the LLM to write the user stories and acceptance criteria for the first two situations listed in the assignment.

Errors: I reviewed the LLM-generated user stories and acceptance criteria against the project requirements and did not identify any errors that required correction.

User Story 1 — Choose a Board Size

User Story Description:
As a human player, I want to choose the size of the Solitaire board so that I can play the game using my preferred board size.

Priority: High
Estimated Effort: 1 hour

Acceptance Criterion 1.1 — Select a valid board size

Given the Solitaire application is open and the available board sizes are displayed,
When the player selects one of the available board sizes,
Then the selected board size is stored as the board size to be used when starting a new game.

Acceptance Criterion 1.2 — Change the selected board size

Given the player has already selected a board size,
When the player selects a different available board size,
Then the newly selected board size replaces the previously selected board size.

Status: toDo

User Story 2 — Choose the Board Type

User Story Description:
As a human player, I want to choose the type of Solitaire board so that I can play using my preferred board configuration.

Priority: High
Estimated Effort: 1 hour

Acceptance Criterion 2.1 — Select a board type

Given the Solitaire application is open and the available board types are displayed,
When the player selects **English, Hexagon, or Diamond**,
Then the selected board type is stored as the board type to be used when starting a new game.

Acceptance Criterion 2.2 — Change the selected board type

Given the player has already selected a board type,
When the player selects a different available board type,
Then the newly selected board type replaces the previously selected board type.

Status: toDo